

Look-Ahead Bias Audit Checklist

Project: Backtesting the ProntoNLP Earnings-Call ATC Signal

Author: Yueqi Lin

Reference: Student Handout §3 — *Look-Ahead Bias: The Audit You Must Pass*

One-Page Sign-Off

#	Item	Pass	One-line justification
3.1	Entry timing (AMC vs BMO)	?	<code>hour < 16 UTC</code> ? next NYSE trading day; <code>hour < 16 UTC</code> ? same day. Vectorised <code>np.searchsorted</code> on SPY-derived calendar. Assertions in <code>00_data_prep.ipynb</code> cell 19.
3.2	Return columns disjoint from features	?	<code>{return_1d...return_20d}</code> ? <code>feature_cols = ?</code> (assertion in cell 25, under the §9b markdown header at cell 24). T2 programmatic test: 0 overlaps / 772 features.
3.3	Cross-sectional features point-in-time	?	All cross-sectional features (sector ranks, Z-scores) deferred to <code>01_analysis.ipynb</code> walk-forward; computed per fold on training-set events only.
3.4	Feature selection inside training	?	IC top-30 sparse selection re-fit on each fold's training data (no full-sample selection).
3.5	Imputation and scaling inside training	?	<code>StandardScaler.fit(train)</code> ? <code>transform(test)</code> pattern inside walk-forward loop.
3.6	Universe membership point-in-time	?	RU3K is fully PIT via CRSP top-3000 mcap proxy (annual June reconstitution; T11 verifies). S&P 500/1500 use current composition with explicit caveat per handout §6.3 fallback path.
3.7	<code>INGESTDATEUTC</code> ≠ availability date	?	Mean lag 1,658 days (batch backfill); entry date uses <code>MOSTIMPORTANTDATEUTC</code> only, decision documented with statistical justification. T6: 0 <code>INGEST</code> features in model input.
3.8	No “future” QoQ deltas	?	All QoQ/2Q/YoY deltas use <code>.shift(1/2/4)</code> within ticker sorted by <code>entry_date</code> ascending — past data only. T8 programmatic verification.
3.9	Corporate-action / delisting handling	?	NaN-return events excluded (no roll-forward). Rule documented in <code>00_data_prep.ipynb</code> §9b and §7 of the research PDF.
3.10	Hyperparameter tuning leaks	?	Initial tuning on 2010–2017 sub-period; walk-forward 2018Q1+ uses frozen hyperparameters. No full-sample grid search.

Programmatic verification: T1–T8 in `02_lookahead_tests.ipynb`, T9–T14 in `06_wrds_lookahead_tests.py`,

and T15–T17 in 07_audit_gap_tests.py — all 17 tests PASS.

Signed: Yueqi Lin · Date: May 2026 · Course: LLM-Driven Quant Research — Final Assignment

Detail (per-item write-ups)

The sections below give the full implementation detail for each audit item. The one-page sign-off above is the deliverable required by handout §2.3; the detail below is provided for thoroughness.

§3.1 — Entry Timing: AMC vs BMO

Rule: Parse MOSTIMPORTANTDATEUTC. If hour $\geq$ 16 UTC $\Rightarrow$ after-market-close $\Rightarrow$ entry is the next trading day's close. If hour < 16 UTC $\Rightarrow$ entry is the same trading day's close.

Implementation: 00_data_prep.ipynb, cells 18–19. Vectorized via `numpy.searchsorted` on the NYSE calendar array derived from SPY prices.

Gray-zone rule (13–16 UTC, documented per §3.1): The handout notes that calls with hour $\geq$ 16 UTC are clearly AMC and hour < 13 UTC are clearly BMO; calls in the 13–15 UTC window (9 AM–11 AM ET, i.e., during regular market hours) are a gray zone. The distribution in cell 18 shows 131,930 events (35%) fall in this 13–15 UTC range. Design choice: treat all calls with hour < 16 UTC as same-day entry (including the 13–15 UTC gray zone). Rationale: these calls occur before the 4 PM ET close; traders receiving the transcript during market hours can act the same day. This is the more conservative (less look-ahead-aggressive) choice — it gives no extra processing time. Using the next day for 13–15 UTC calls would be an acceptable alternative but would waste one day of alpha. Rule is documented here and in cell 18 of 00_data_prep.ipynb.

Assertions added (cell 19): - `entry_date > call_date` for all AMC events (hour $\geq$ 16 UTC) $\Rightarrow$ - `entry_date >= call_date` for all BMO/gray-zone events (hour < 16 UTC) $\Rightarrow$

Status: $\Rightarrow$ PASS

§3.2 — Forward Returns Are Targets, Never Inputs

Rule: `return_Nd` columns are computed *from* entry price. They must never appear in the model feature set.

Implementation: Returns computed in cells 22–23 after `entry_date` is locked. Feature columns built in cells 29–30 from `AspectTheme` matrix and `EventScores` only.

Assertion added (cell 25, immediately after the §9b markdown header at cell 24): - `{return_1d, return_3d, return_5d, return_10d, return_20d}` is disjoint from `feature_cols` $\Rightarrow$

Status: $\Rightarrow$ PASS

§3.3 — Cross-Sectional Features Must Be Point-in-Time

Rule: Z-scores, percentile ranks, sector means computed across “all events in the same quarter” leak future data for events early in the quarter. Use only events that have already happened (expanding-window percentile).

Implementation: Cross-sectional features (sector ranks, Z-scores) are deferred to `01_analysis.ipynb` and computed per walk-forward fold using only training-set events. No cross-sectional transformations are applied in `00_data_prep.ipynb`.

Status:  PASS (deferred by design; documented in cell 0 markdown)

§3.4 — Feature Selection Is Part of Training

Rule: Spearman/mutual-info ranking, PCA fits, target-encoding must be done on training fold only, refitted at every walk-forward step.

Implementation: No feature selection is performed in `00_data_prep.ipynb`. All selection happens inside the walk-forward loop in `01_analysis.ipynb`, fit on train set, applied to test set.

Programmatic backstop: T15 in `07_audit_gap_tests.py` re-runs the IC top-30 selector on two disjoint training windows and asserts the selected feature sets are data-dependent (Jaccard  0.05), confirming selection cannot be a constant computed on the full sample.

Status:  PASS (deferred to notebook 01 by design; T15 verifies data-dependence)

§3.5 — Imputation and Scaling Are Part of Training

Rule: `StandardScaler.fit` and median imputation must use training data only. Pattern: `fit_transform(train)`  `transform(test)`.

Implementation: No scaling or imputation in `00_data_prep.ipynb`. Applied inside walk-forward loop in `01_analysis.ipynb`.

Status:  PASS (deferred to notebook 01 by design)

§3.6 — Universe Membership Is Point-in-Time

Rule: Today’s S&P 500 is not 2014’s S&P 500. Use historical constituents or document the survivorship-bias caveat.

Implementation: - Russell 3000 is point-in-time. WRDS / CRSP provides a survivorship-free proxy: top-3000 US common stocks by market capitalisation, snapshotted at each annual June reconstitution (matches Russell methodology). Implemented in `03_wrds_pull.py` (CRSP msf market-cap pull, June snapshots) and `04_wrds_integrate.py` (`in_RU3K_PIT` flag built via `merge_asof` of event `entry_date` against the snapshot windows). `06_wrds_lookahead_tests.py` T11 verifies that every `in_RU3K_PIT=True` event has `snap_date < entry_date` (zero violations / 153,988 events). - S&P 500 and S&P 1500 use current (2026) composition. Historical removed members are not available

in the Compustat subscription tier accessed (`comp.idxcst_his` returned only `thru_date = NULL` rows). Per handout §6.3’s fallback path, the survivorship-bias caveat is documented explicitly in: - `00_data_prep.ipynb` cell 10 markdown - `reports/research_report.md` §7 (Risks and Limitations) - `reports/research_report.md` §2.4 / §2.5 (universe and price-data definitions) - `README.md` Key Design Decisions

Reported S&P alpha should be treated as an upper bound. Russell 3000 alpha is survivorship-free in both universe assignment and price coverage (CRSP-first preference; see research report §2.4).

Status:  PASS (RU3K is fully PIT; S&P caveat explicitly documented per handout §6.3)

§3.7 — INGESTDATEUTC ≠ Availability Date

Rule: Some calls were ingested by ProntoNLP days after the actual call. For the strictest backtest, entry date = $\max(\text{MOSTIMPORTANTDATEUTC} + \text{entry-rule}, \text{INGESTDATEUTC})$. Document whichever rule you choose.

Finding: Inspection of `INGESTDATEUTC` (cell 18) reveals a mean lag of 1,658 days (4.5 years) with 83.7% of events having a positive lag. This confirms the field records a batch historical backfill — ProntoNLP processed the full historical archive in bulk, not in real time. Applying this constraint would push 81% of entry dates to 2023+, joining 2010 signals to 2023 prices — which is itself severe look-ahead.

Design choice: Entry date uses `MOSTIMPORTANTDATEUTC` only. `INGESTDATEUTC` is parsed and the lag distribution is printed in cell 18 for transparency.

Status:  PASS (choice documented with statistical justification)

§3.8 — No “Future” QoQ Deltas

Rule: QoQ features (current quarter minus previous quarter) are fine. “Next quarter minus current” is not.

Implementation: QoQ deltas computed in cell 30 via `groupby('BESTTICKER')[cols].shift(1)` on data sorted by `entry_date` ascending. Each event’s delta uses the previous event for that ticker only.

Status:  PASS

§3.9 — Corporate-Action / Delisting Handling

Rule: Don’t assume a fill on a non-tradable day. If price source returns NaN, skip the trade or roll forward. Document the rule.

Implementation: Forward returns are joined via left-merge on (`BESTTICKER`, `entry_date`). Events with no price data receive NaN returns and are excluded from IC/quintile analysis (not forward-filled or imputed). Events where the exit price is missing (ticker delisted before exit date) also receive NaN. No roll-forward is applied.

Rule documented: NaN-return events are excluded from all analyses. This is conservative and avoids assuming liquidity.

Programmatic backstop: T16 in 07_audit_gap_tests.py greps 00_data_prep.ipynb and 01_analysis.ipynb for any fillna(0) / bfill / ffill applied to return_* columns and asserts none are present, so a delisted ticker can never be silently treated as a zero-return trade.

Status:  PASS (NaN-exclusion rule documented; T16 verifies no mass-fill on returns)

§3.10 — Hyperparameter Tuning Leaks Too

Rule: If you grid-search on the full sample and re-run the backtest with the winners, you have leaked. Tune on a held-out sub-period (e.g., 2010–2017), then freeze and walk forward from 2018+.

Implementation: Hyperparameter tuning is performed in 01_analysis.ipynb using only the training portion of each walk-forward fold. The initial tuning sub-period is 2010–2017; walk-forward test begins 2018Q1. No full-sample grid search is performed.

Programmatic backstop: T17 in 07_audit_gap_tests.py greps 01_analysis.ipynb for GridSearchCV, RandomizedSearchCV, optuna, and hyperopt and asserts none are imported or invoked, ruling out any full-sample tuning library call.

Status:  PASS (deferred to notebook 01 by design; T17 verifies no tuning library is invoked)

Summary

#	Item	Status
3.1	Entry timing (AMC/BMO)	 PASS
3.2	Forward returns as targets only	 PASS
3.3	Cross-sectional features	 PASS
	point-in-time	
3.4	Feature selection in training only	 PASS
3.5	Imputation/scaling in training only	 PASS
3.6	Universe membership	 PASS (RU3K is fully PIT via CRSP; S&P caveat documented per handout §6.3)
	point-in-time	
3.7	INGESTDATEUTC $\neq$ availability date	 PASS (choice documented)
3.8	No future QoQ deltas	 PASS
3.9	Corporate-action / delisting handling	 PASS
3.10	Hyperparameter tuning	 PASS

All 10 look-ahead audit items: PASS

Sign-Off

I certify that all 10 look-ahead audit items above have been reviewed and pass. The backtest implementation in `00_data_prep.ipynb` and `01_analysis.ipynb` contains no look-ahead bias that I am aware of, and all 17 formal bias tests pass programmatically: T1–T8 in `02_lookahead_tests.ipynb`, T9–T14 in `06_wrds_lookahead_tests.py`, and T15–T17 in `07_audit_gap_tests.py`.

Signed: Yueqi Lin

Date: May 2026

Course: LLM-Driven Quant Research — Final Assignment